

Case:

The Global Financial Crisis of 2007–2009

We begin our look at the 2007–2009 financial crisis by examining three central factors:

1. Financial innovation in mortgage markets
2. Principle-Agent problems in mortgage markets
 1. Mortgage brokers
 2. Borrowers
 3. Commercial banks, investment banks and insurance companies
3. The role of asymmetric information in the credit rating process

2. Agency problems in mortgage markets

Profit vs. Risk control: Which one do mortgage brokers sacrifice?

1. Mortgage brokers

- Due to the invention of mortgage-backed securities, mortgage brokers
 1. Issue loan to borrowers
 2. Securitization them
 3. Quickly sell mortgage-backed securities to investors
 - What's the potential problem?
 - Principal-agent problem
 - Mortgage brokers earned fees from the volume of the loans produced, not the quality
- Brokers encourage households to take on mortgage aggressively.

2. Agency problems in mortgage markets

Agency problems exacerbate :

Moral hazard occurs when one party engages in riskier behavior because they do not bear the full consequences of their actions.

2. Borrowers

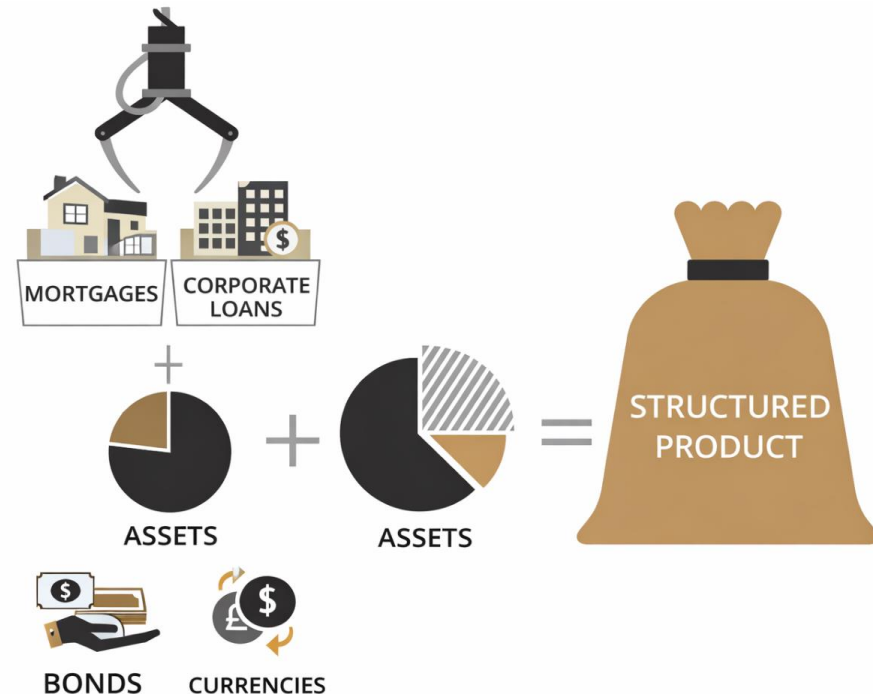
- Become risk-taking investors, expecting housing prices went up; if went down, they “walk away”.
- Unqualified borrowers even bought houses they could not afford through either creative mortgage products or fraud
- i.e. Credit risk ↑ ↑
- Become risk-taking investors
 - So banks need to be heavily regulated

2. Agency problems in mortgage markets

Agency problems exacerbate :

3. Commercial banks, investment banks and insurance companies

- Invents complicated structured products
- High return through securitization
- Weak incentive to control risk



Summary: Agency problems in mortgage markets

1. Financial innovation in mortgage markets
2. Agency problems in mortgage markets
 - Mortgage brokers
 - Issue mortgage loan to borrowers
 - Quickly sell MBS to investors
 - Borrowers
 - Excessive borrowing
 - Many unqualified borrowers
 - Commercial banks, investment banks and insurance companies
 - Invention of structured products
 - Earn a lot through securitization

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3. The role of asymmetric information in the credit rating process

3. The role of asymmetric information in the credit rating process

- Process
 - **Investment banks create structured financial products** and approach CRAs for ratings. (Banks pay for this service)
 - **CRAs analyze the risk** and assign a rating based on expected default probability.
 - **Investment banks ask for guidance** on how to improve the structure of the product to receive a higher rating. (Banks pay)
 - **CRAs suggest modifications** (e.g., changing the mix of assets, or adjusting tranche structures).
 - **Investment banks restructure the product** accordingly and resubmit for a new rating.

3. The role of asymmetric information in the credit rating process

- **Adverse selection issue**
- Investment banks assemble structured products by pooling assets (e.g., mortgages, corporate loans).
- Who knows the true risk of the product?
 - Investment banks have better knowledge of it → Adverse selection Issue?
- Investment banks may include lower-quality assets while structuring the products in a way that still achieves a high rating. Difficult for CRAs to find it out?
 - CRAs, relying on historical models and assumptions rather than real-time data, may not fully account for hidden risks.
- Who suffer?
 - Investors, trusting the high ratings, unknowingly purchase products with more risk than they perceive.

3. The role of asymmetric information in the credit rating process

- **Moral Hazard**
- Inflate the rating v.s. Focus on quality?
 - Investment banks manipulate structuring to secure the highest ratings.
- Why CRAs are lack of incentive to provide unbiased rating?
 - **Profit from fees paid by investment banks**, want to maintain business relationships with investment banks.
- Who suffer? investment banks? CRAs? Investors?
 - Investment banks pass the risk to investors and do not suffer direct losses if defaults occur.
 - Investors, underestimate risk, overinvest in toxic assets.

Summary: Asymmetric information in the credit rating process

- What could be the asymmetric information problems?
- **Adverse Selection**
 - Banks hide poor-quality assets in structured products, which still receive high ratings. Investors unknowingly buy riskier securities.
- **Moral Hazard**
 - Banks take excessive risks, knowing they can pass the risk to investors and get high ratings from CRAs who have incentives to maintain their business relationships.
 - Moral hazard occurs when one party engages in riskier behavior because they do not bear the full consequences of their actions.

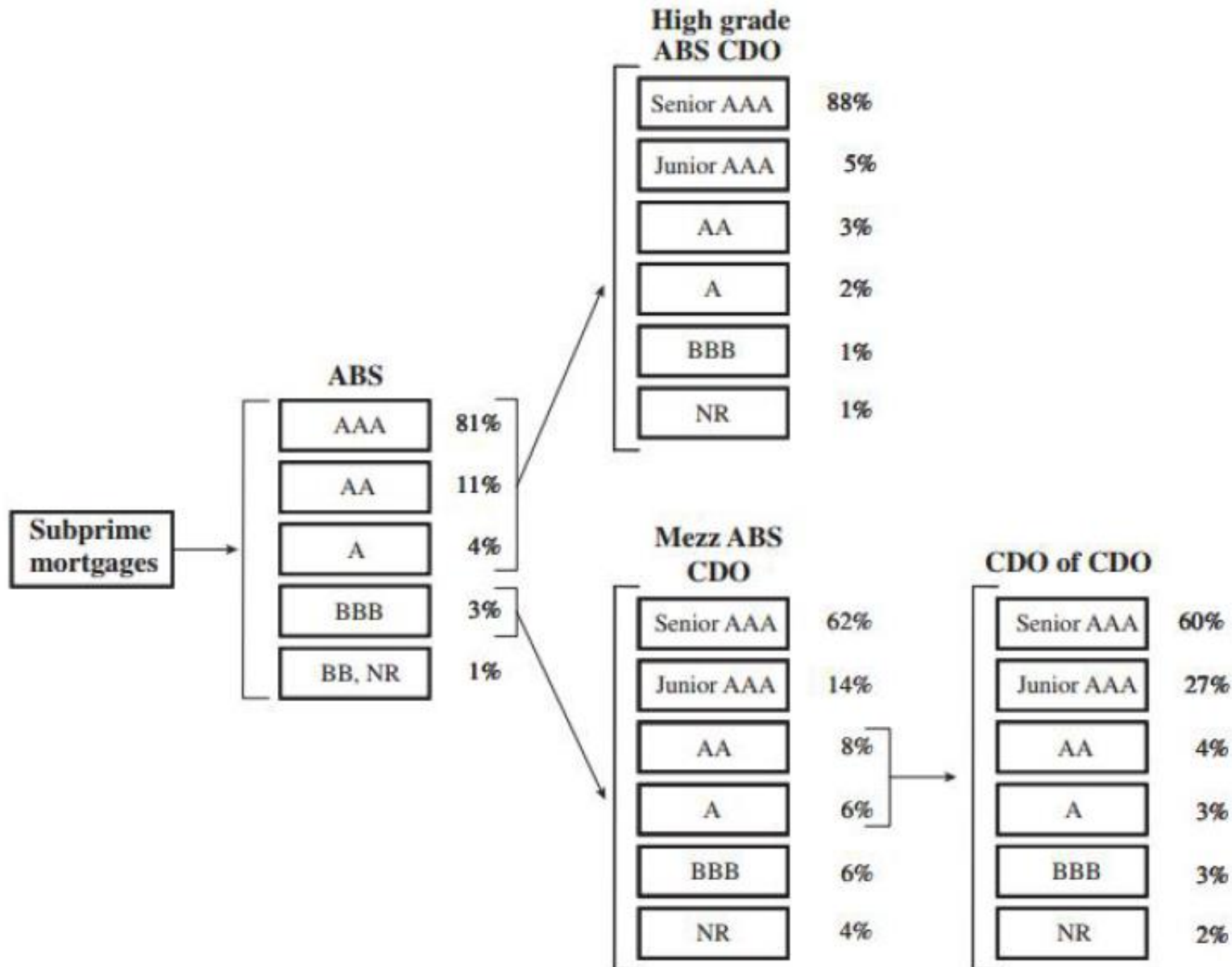
Mini-Case: CDOs (1 of 3)

An example of financial innovation

Collateralized Debt Obligations (CDOs) : A structured product
SPV and CDOs

- *Special purpose vehicle (SPV):*
 - A corporate entity which buy assets (e.g. corporate bonds, mortgage-backed securities)
 - create securities from those assets
 - sell those securities to investors.
- In a CDO, the securities (or tranches) are created based on default priorities. The first defaults go to the lowest rated tranches. The highest rated tranches suffer defaults if most of the assets default.

Mini-Case: CDOs (1 of 3)



Mini-Case: CDOs (2 of 3)

There are many, many tranches in a CDO, each with different exposure to defaults:

- The highest rated tranches are called super senior tranches
- The next bucket is known as the senior tranche – it has a little more risk and pays a higher interest rate
- The next tranche is the mezzanine tranche - it bears more risk and has an even higher interest
- The lowest tranche is the equity tranche - this is the first tranche that suffers losses from defaults

Mini-Case: CDOs (3 of 3)

“(I) would like to know what those damn things are worth.”
——Ben Bernanke, the chairman of the Federal Reserve

- Structured products are complicated!
 - Difficult to determine what they are worth and who has the rights to what cash flows.
 - There are even complexed CDO², CDO³...
 - Financial engineering contribute to financial innovation, here are the dark sides.
- Increased complexity of structured products lead to...
- → Useful information ?
- → Adverse selection and moral hazard ?

Summary:

3 factors in the Financial Crisis of 2007–2009

1. Financial innovation in mortgage markets
 - Financial derivatives
2. Agency problems in mortgage markets
 - Mortgage brokers
 - Borrowers
 - Commercial banks, investment banks and insurance companies
3. Asymmetric information in the credit rating process
 - May be credit rating agencies need to be rated

Effects of the Financial Crisis 2007-2009

Many suffered as a result of the 2007–2009 financial crisis.
We will look at five areas:

1. U.S. residential housing: Bubble bursts
2. FIs balance sheets
3. The “shadow” banking system
4. Global financial markets
5. The failure of major financial firms

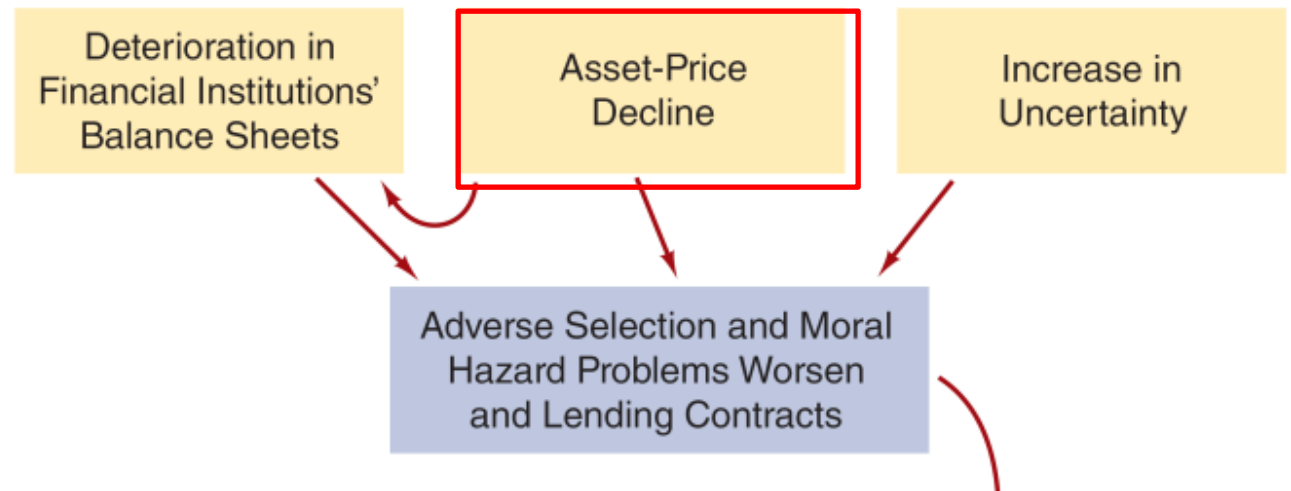
Recall

Stage One: Initial Phase

Financial crisis can begin in several ways:

1. Credit Boom and Bust
- 2. Asset-Price Boom and bust**
3. Increase in Uncertainty

STAGE ONE
Initial
Phase



Effects of the Financial Crisis 2007-2009

1. Residential housing

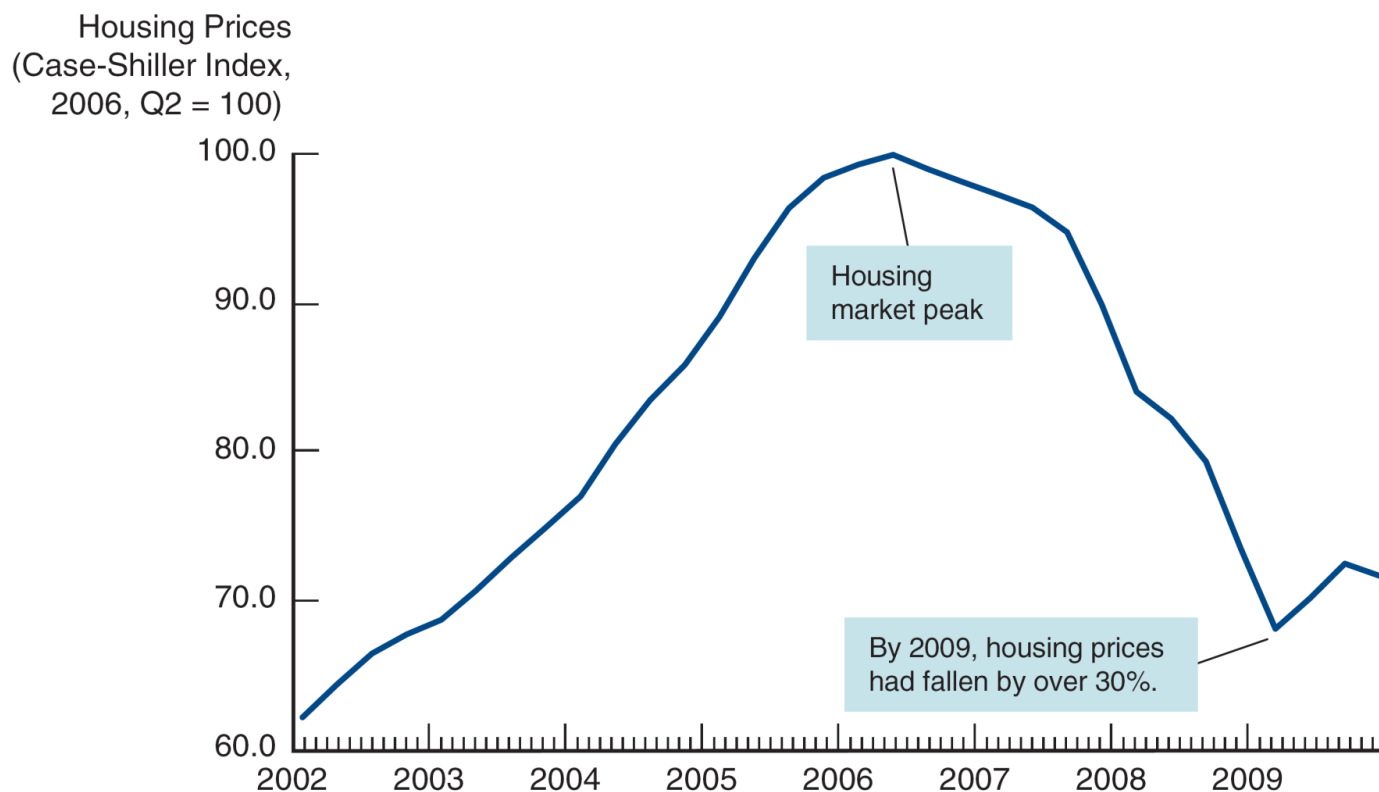
- The housing boom was lauded by economics and politicians, initially.
 - dotcom Tech bubble in 1995-2000
 - Led to a “democratization of credit”
 - Raise U.S. homeownership rate
- The housing boom stimulate growth in the subprime market.
- People buy houses even they do not have money. How can they get the mortgage?
- Underwriting standard fell.

Effects of the Financial Crisis 2007-2009

Residential housing

- Why mortgage originators approve mortgage application from poor-credit borrower?
- Huge profit
- Afraid of borrowers' default?
 - As long as the housing price is going up, banks gain
- Borrowers are not worried?
 - If housing price goes up... 😊
 - If housing price goes down... 😐
- i.e. They do not bear the full consequences of their actions.

Figure 4 Housing Prices and the Financial Crisis of 2007–2009



Source: Case-Shiller 20-City Composite Home Price Index in Federal Reserve Bank of St. Louis, FRED database:
<https://fred.stlouisfed.org/series/SPCS20RSA>.

Why boom and bust in the housing market?

- Why boom?
 - Low interest rate on residential mortgages.
- How?
 - Federal Reserve monetary policy- lower interest rates
 - Fannie Mae and Freddie Mac purchase trillions of dollars of mortgage-backed securities
 - Huge cash flooded into U.S. from countries like China and India.

* Fannie Mae (Federal National Mortgage Association, FNMA) and Freddie Mac (Federal Home Loan Mortgage Corporation, FHLMC) are government-sponsored enterprises in the U.S.

Was the Fed to Blame for the Housing Price Bubble? (1 of 2)

- Some argue that low interest rates from 2003 to 2006 fueled the housing bubble (the Taylor rule).
- In early 2010, Mr. Bernanke rebutted this argument. He argued rates were appropriate.



<https://www.youtube.com/watch?v=zOKCw3h4Wjs>

Was the Fed to Blame for the Housing Price Bubble? (2 of 2)

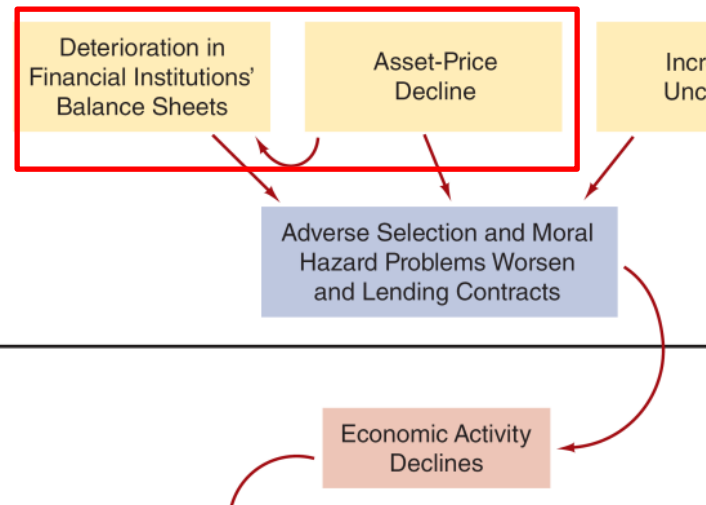
- He also pointed to new mortgage products, relaxed lending standards, and capital inflows as more likely causes.
- Bernanke's speech was very controversial, and the debate over whether monetary policy was to blame for the housing price bubble continues to this day.

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STAGE ONE
Initial
Phase



STAGE TWO
Banking
Crisis

Case: The Global Financial Crisis of 2007-2009

2. Deterioration of Financial Institutions' Balance Sheets

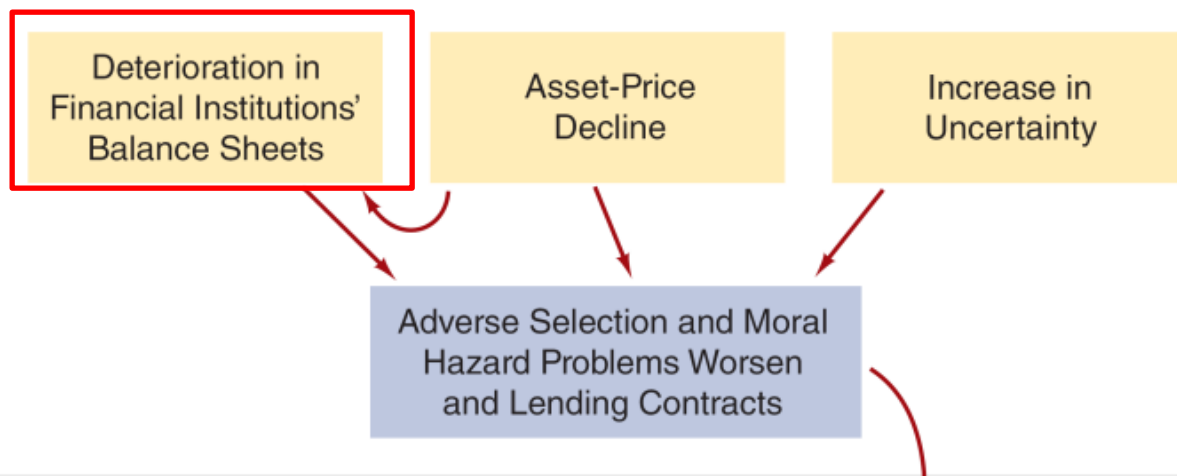
- As mortgage defaults rose, value of assets fall in banks and other FIs.
 - This was further complicated by the complexity of mortgages, CDOs, defaults swaps, and other difficult-to-value assets.
- How do banks net worth change?
- What could banks do?
 - Deleveraging, selling assets and restricting credit
- What could be the consequence for the economic activities?
 - Limited the loan issuing service → few of them collect information for investment opportunities → information asymmetry → further depressing the struggling economy

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STAGE ONE
Initial
Phase



Case: The Global Financial Crisis of 2007-2009

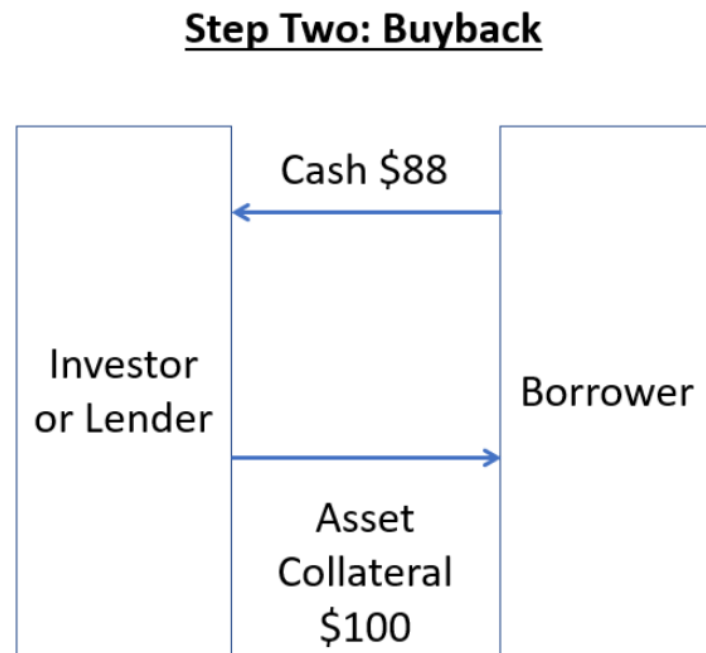
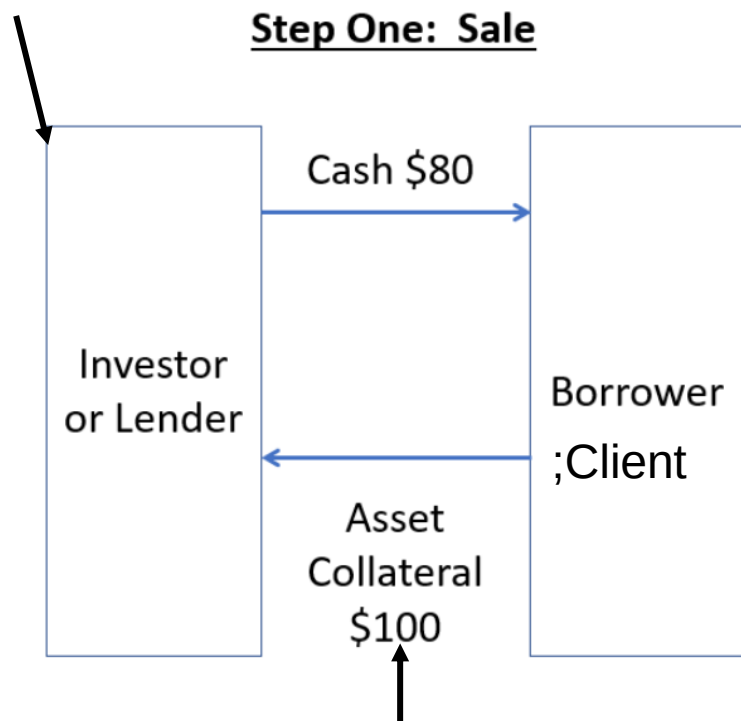
3. Run on the Shadow Banking System

- The shadow banking system also experienced a run, when housing bubble burst, why?
- Shadow banks: hedge funds, investment banks, and other liquidity providers in our financial system
 - Provide funds for investors that cannot get fund from banks.
- Where do shadow banks get funds?
 - Repurchase agreements (repos), not deposits
- What is repos?
 - Use assets like mortgage-backed securities as collateral, to borrow money.

Repurchase agreement (Repos)

- A repo is economically similar to a secured loan, with the lender receiving securities with collateral to protect himself against default by the seller.

Large institutional investors
(eg. Mutual funds, Hedge fund)



Eg. Mortgage-backed securities

Repurchase agreement (Repos)

- Repurchase agreement (similar to a secured loan)
- The repo market is an important source of funds for large financial institutions in the non-depository banking sector
- Which type of bank tend to collapse first?
 1. Rely on **deposits** for its funding
 2. Rely on **repos** for its funding

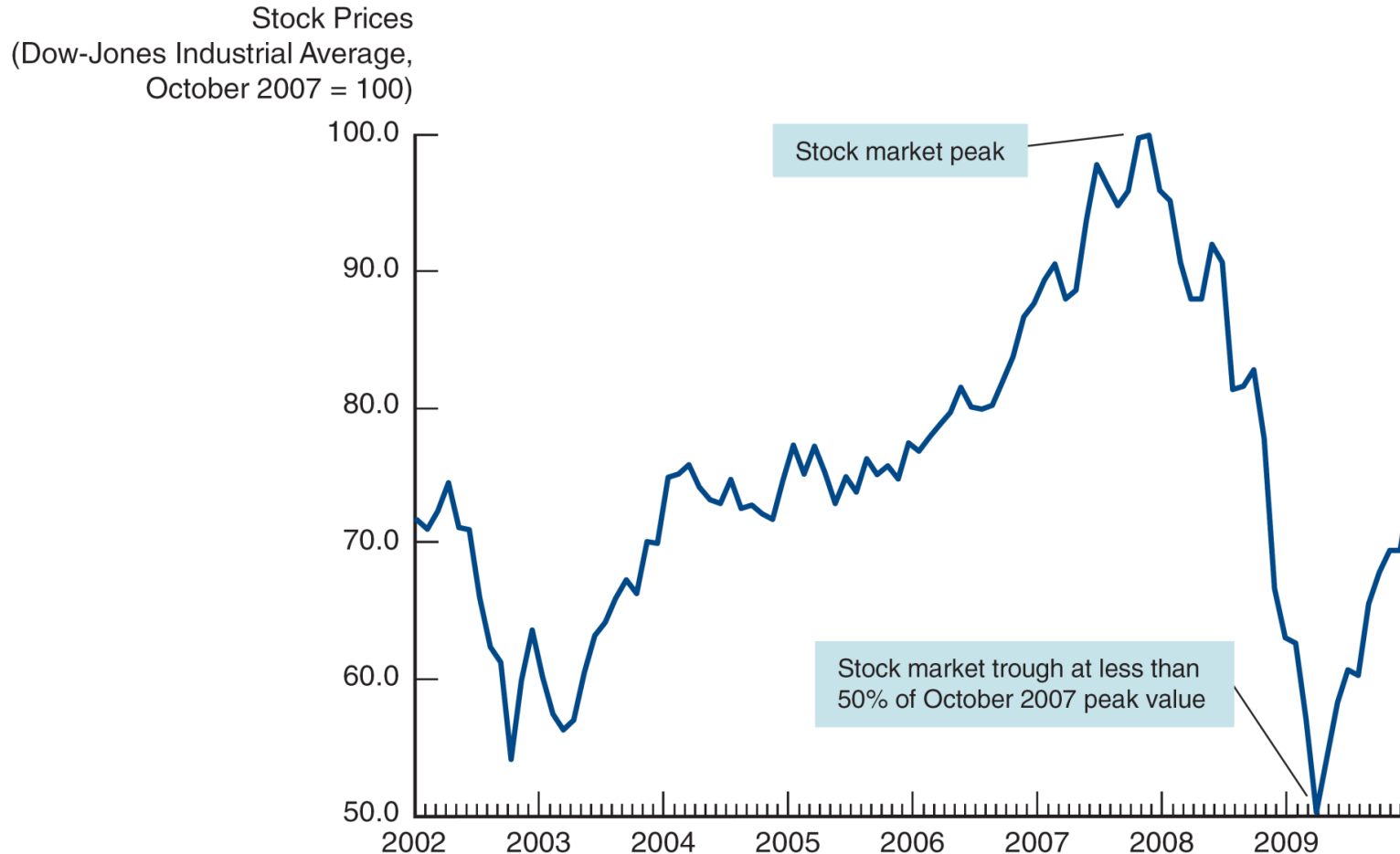
Case: The Global Financial Crisis of 2007-2009

3. Run on the Shadow Banking System

- The rising defaults on mortgages →
- Increased or decreased fund in shadow banks?.
- What can shadow banks do to raise more fund?
 - Fire sales their assets, including mortgage-backed securities.
- What are the results for fire sales?
 - Shadow banks collaterals worth even less
 - They're trapped in a downward circle

Figure 5 Stock Prices and the Financial Crisis of 2007–2009

Decline in asset prices in the stock market (which fell by over 50% from Oct. 2007 to Mar.2009)



Along with the fire sales from the run on the shadow banking system, weakened both firms' and households' balance sheets.

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Case: The Global Financial Crisis of 2007-2009

Global contagion effect

- Europe was actually first to raise the alarm in the crisis.
- With the downgrade of \$10 billion in mortgage related products, short term money markets froze.
 - BNP Paribas froze 1.6 billion euros worth of funds, barring investors from redeeming cash from the funds, in August 2007.
 - The run on the shadow banking system began.



Case: The Global Financial Crisis of 2007-2009

- Injections of liquidity from central banks, but quickly drying up.
- Leading to several bank failures.
- By most standards, Europe experienced a more severe downturn than the U.S.



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Global: The European Sovereign Debt Crisis

- Global: The European Sovereign Debt Crisis
- During the financial crisis, European countries suffer from economic contraction:
 - Lower tax revenue from economic contraction
 - High expenses for FI bailouts
 - Fear of gov't default
 - Interest rates surge
- Greece was the first domino to fall

Global: The European Sovereign Debt Crisis

- Global: The European Sovereign Debt Crisis
- In October 2008, Greek gov't debt-to-GDP =160%
- Interest rate on Greek debt : 40%
 - 4% in year 2006

Global: The European Sovereign Debt Crisis

- Greece was forced to write-down its debt (partial default)
 - Despite of bailouts from other European countries and European Central Bank
 - The prime minister was eventually forced to resign.
- Spread to other European countries.
- Gov't forced to take austerity measures.
- Interest rate soared to double-digit level.
- Unemployment rates climbed
 - Unemployment rate in Spain:25%
- Civil unrest broke out

Case: The Global Financial Crisis of 2007-2009

Collapse of Big investment firms in the U.S.

- Bear Sterns –sold to →JP Morgan, for only 5% of its value
- Lehman Brothers files for bankruptcy.
 - the largest bankruptcy filing in U.S. history
- Merrill Lynch –sold to → Bank of America at “fire” sale prices, 40% of its value.
- Government bailed out AIG.

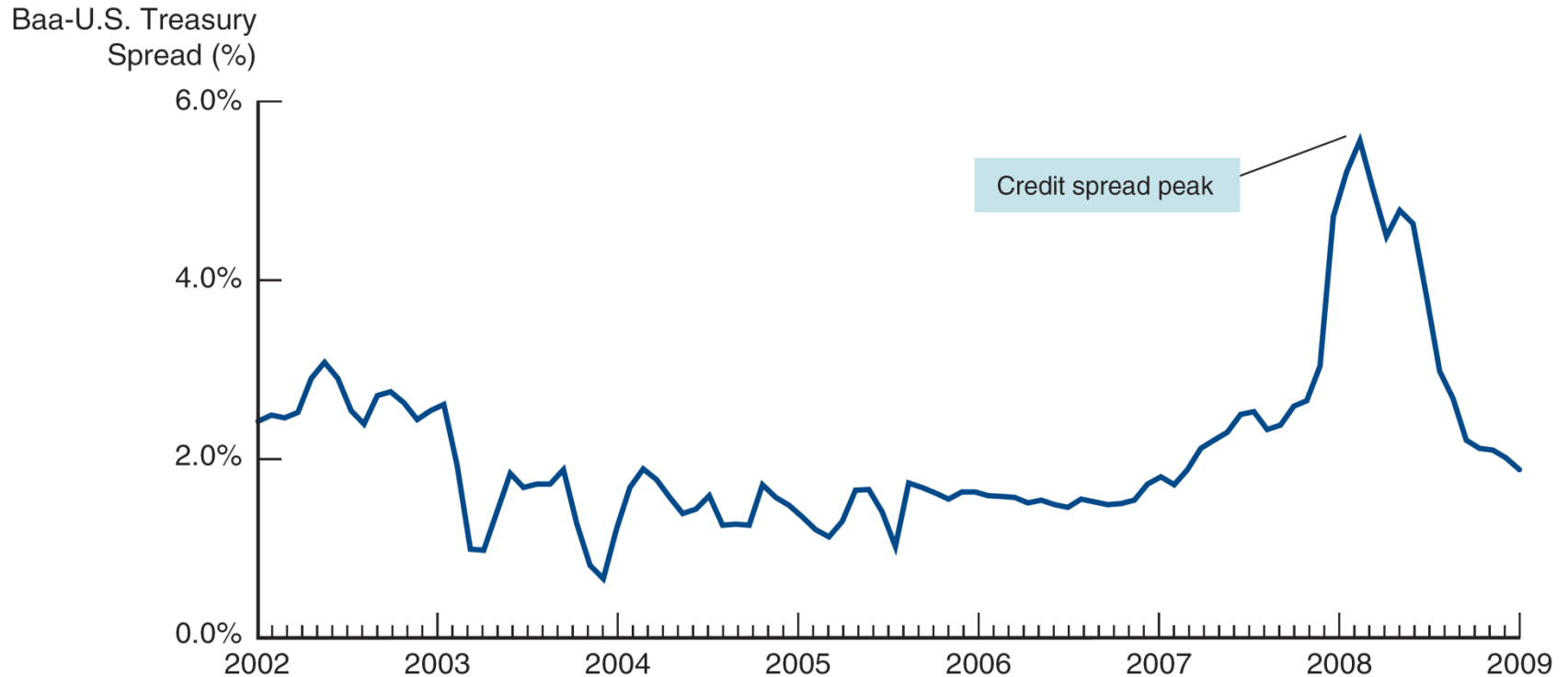


Case: The Global Financial Crisis of 2007-2009

The crisis and impaired credit markets have caused the worst economic contraction since World War II.

- The crisis peaked in September of 2008.
- Congress passed a bailout package, but the stock market continued to decline, and credit spreads reached over 500 bps.
- Real GDP was -1.3% and unemployment rate was over 10%.

Figure 6 Credit Spreads and the 2007–2009 Financial Crisis



Source: Federal Reserve Bank of St. Louis, FRED database, <https://fred.stlouisfed.org/series/BAA10Y>.

Case: The Global Financial Crisis of 2007-2009

- Starting in March 2009, a bull market in stocks got under way and credit spreads began to fall.
- Unfortunately, the pace of the recovery has been slow.

Chapter Summary

- What happens during Financial Crises?
- A large disruption to information flows in financial markets
 - Financial frictions and credit spreads↑
 - Financial markets incapable of channeling funds
 - A sharp contraction in economic activity
- Dynamics of Financial Crises in Advanced Economies
 - We examined the stages of a crisis in an advanced economy.
 - We further examined the 2007–2009 U.S. Financial Crisis.

Chapter Summary

- Dynamics of Financial Crises

Stage 1: Credit bubble or asset-price bubble. Excessive risk-taking in FIs

→ Bubble bust or unsustainability in FIs

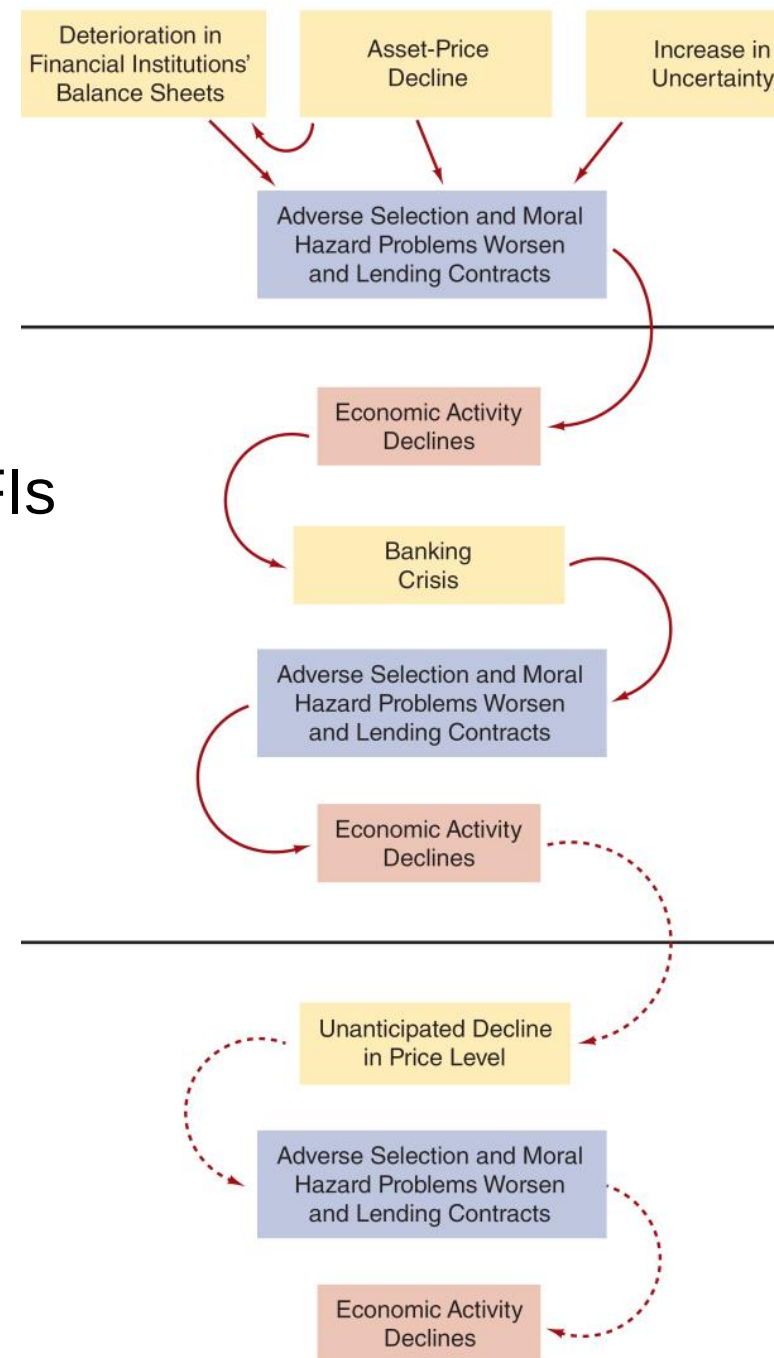
(Worsened adverse selection and moral hazard problems)

→ **A contraction of lending and economy activity**

→ Trigger Stag 2 of Banking crisis

→ **Worsened adverse selection and moral hazard problem**

→ **A further decline of lending and economic activity**



Chapter Summary

- Dynamics of Financial Crises

Economic downturn can trigger Stage 3

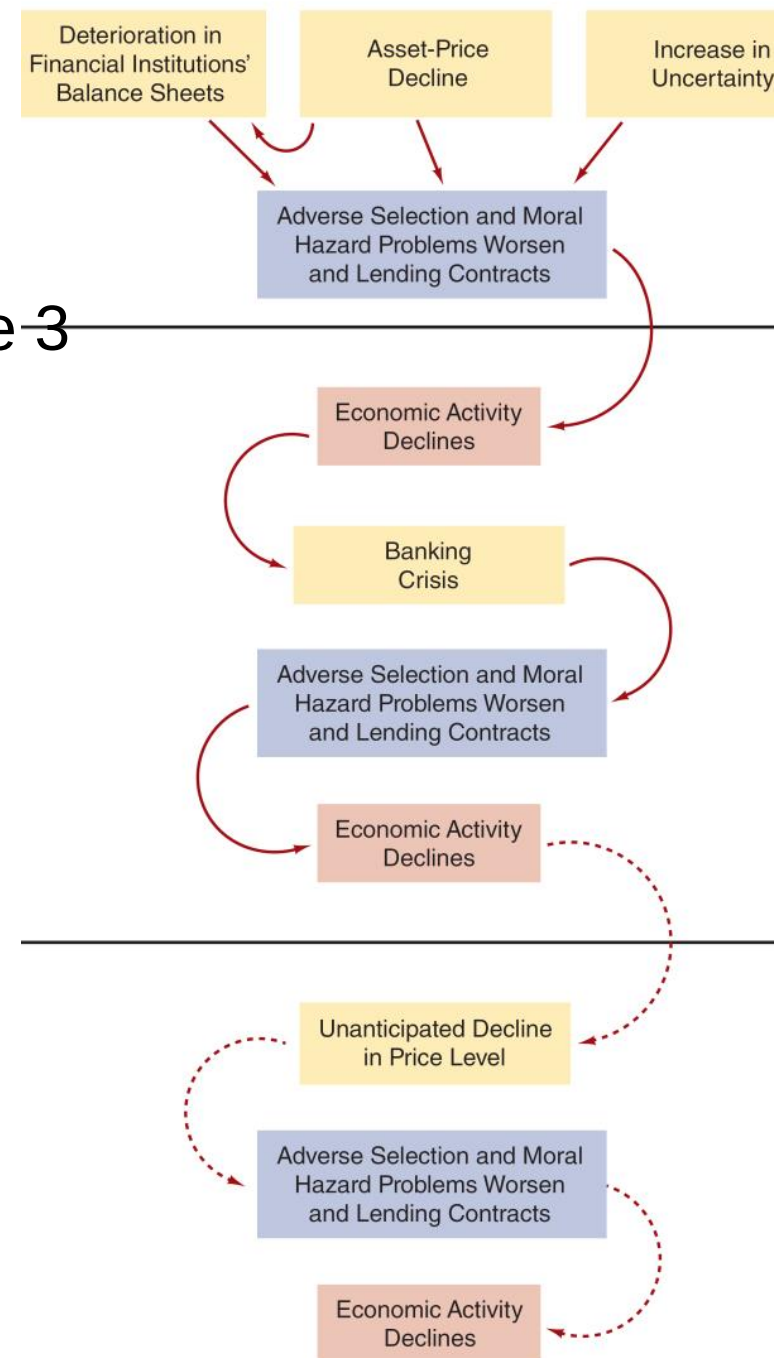
→ Prices plunge

→ Debt deflation

→ Worsened adverse selection and moral hazard problem

→ A further decline of lending and economic activity

- **Downward spiral!**



Chapter Summary

- Huge loss:
- Economic activity remain depressed for a long time
- Large job losses
- Social unrest

